

Md. Maruf Mullah

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ACADEMIC CREDENTIALS

B.Sc. in Mechanical Engineering, Military Institute of Science and Technology (MIST)
CGPA: 3.23 / 4.00

Apr 2021 – May 2025

RESEARCH INTERESTS

- Advanced and Smart Manufacturing Systems
- Computational Mechanics and Numerical Modelling
- Data-Driven and Machine Learning Applications in Engineering
- Advanced Engineering Materials and Material Characterization

BACHELOR'S THESIS

Densification of Natural Wood to Improve Structural Properties

Advisor: Major Md. Anisur Rahman, EME, Assistant Professor, ME Dept., MIST

- Studied structural performance of natural, seasoned, and chemically treated–densified wood.
- Investigated three tropical species: *Swietenia macrophylla*, *Albizia procera*, and *Cordia subcordata*.
- Chemical treatment involved NaOH and Na₂SO₃ followed by thermo-mechanical hot-press densification.
- Conducted mechanical, moisture absorption, and SEM analyses following ASTM standards; densified samples showed highest strength.

PROJECTS

Traditional Machine Learning

Surface Roughness Prediction

2025

- Built regression models (LR, RF, GB, SVM, DT) to predict surface roughness.
- Best model: Decision Tree ($R^2 = 0.85$); analyzed influence of cutting parameters.
- [GitHub](#)

Time-Series Forecasting

Wind Speed & Direction Prediction

2024

- Applied ANN, RNN, LSTM, and CatBoost on meteorological time-series data.
- Performed feature engineering, temporal modeling, and comparative evaluation.
- [GitHub](#)

Deep Learning & Computer Vision

Casting Defect Classification

2026

- Compared CNN, MobileNetV3, ViT, and YOLO classifiers on industrial datasets.
- Achieved 99.9% accuracy using YOLO-based classifier.
- [GitHub](#)

YOLO Vision Applications

2026

- Implemented segmentation (brain MRI), detection (PPE compliance), classification, and real-time tracking.
- Managed dataset preparation and annotation using Roboflow.
- Optimized training pipelines and inference for real-world deployment.
- [GitHub](#)

LLM Applications

RAG Research Assistant

2026

- Developed a local Retrieval-Augmented Generation (RAG) chatbot for research assistance.
- Integrated document ingestion, web scraping, and structured Markdown knowledge storage.
- Enabled local inference and Gemini 2.5 Flash Lite integration for reliable responses.
- Applied in surface roughness research for fast literature insight extraction.
- [GitHub](#)

PUBLICATIONS

Conference Papers and Book Chapters

- A. Tahsin, **Md. Maruf Mullah**, et al. *Impact Strength and Moisture Behaviour of Natural, Densified and Seasoned Wood*. ICMEAS 2025 (Accepted).
- Z. Imtiaz et al., **Md. Maruf Mullah**. *Meteorological Drought Prediction Using Forecasting Models*. ICWFM 2025 (Book Chapter – Accepted).

- I. Alam et al., **Md. Maruf Mullah**. *Effect of Groove Shapes on Microstructural and Mechanical Behavior of Pipe Welds under Post-Weld Heat Treatment*. Manuscript under revision for journal resubmission.

TECHNICAL SKILLS

Programming: Python, C, C++, MATLAB, HTML
Machine Learning: Scikit-learn, NumPy, Pandas, MLflow (Experiment Tracking), Optuna (Hyperparameter Optimization)
Deep Learning & Computer Vision: PyTorch, TensorFlow, YOLO (Detection, Segmentation, Classification, Tracking), Roboflow
LLM & Generative AI: RAG, LLM Inference (Local & API-based), Hugging Face Transformers
Data Visualization: Matplotlib, Seaborn, Excel, Google Sheets
Simulation: ANSYS, COMSOL Multiphysics
CAD / 3D Printing: SolidWorks, FreeCAD, Anycubic Cobra
Documentation: $\LaTeX$, Microsoft Word, Google Docs

PROFESSIONAL EXPERIENCE

Management Trainee Officer (Engineer) — Assessment Program
PRAN-RFL Group, Kaliganj Agro-Processing Ltd. Jul 2025
Kaliganj, Gazipur, Bangladesh (On-site)
• Participated in structured assessment for manufacturing operations and engineering roles.
• Observed production workflow, plant layout, and assisted in machine monitoring.
• Prepared technical and operational reports demonstrating analytical and communication skills.

Mechanical Engineering Intern
IFAD Autos PLC Feb 2024 – Mar 2024
Dhamrai Upazila, Dhaka, Bangladesh (On-site)
• Assisted in vehicle assembly, mechanical fitting, and torque verification.
• Observed quality control procedures and assembly line balancing.
• Gained hands-on exposure to automotive manufacturing systems.

CO-CURRICULAR ACTIVITIES & AWARDS

Government Scholarships
Recipient of Government Scholarship in the **Primary Education Completion (PEC)** Examination 2012
Recipient of Government Scholarship in the **Junior School Certificate (JSC)** Examination 2015
Science Fair Champion — Upazila Level, Raipura 2017
Led a school team to first place by presenting an applied science project.

Bangladesh Boy Scouts — Scout Volunteer 2015–2016
Served as a scout volunteer at high school, supporting discipline management, event organization, and community service activities.

Volunteer, ICMEAS 2022
Assisted in conference logistics, participant coordination, and technical session management.

Volunteer, Job Fair 2022
Supported employer coordination, registration, and event operations.

Organizing Committee Member, Soccer Bot Competition — MIST 2024
Contributed to planning, team coordination, and competition execution under MIST Robotics initiatives.

Humanitarian Volunteer 2024
Engaged in flood relief operations and winter clothing distribution programs.

CERTIFICATIONS

- Machine Learning Specialization (Supervised Learning) — Coursera 2024
- Neural Networks and Deep Learning (Neural Networks and Deep Learning) — Coursera 2025
- Mathematics for Data Science — Simplilearn 2025
- Python Programming Bootcamp — Decoders Academy 2025
- MATLAB Onramp — MathWorks 2025
- Git Training — Simplilearn 2025
- Google Sheets — Simplilearn 2025
- Project-Based Excel — Grameenphone Academy 2025

REFERENCES

Major Md. Anisur Rahman, EME
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Dr. Md. Sayem Hossain Bhuiyan
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